AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

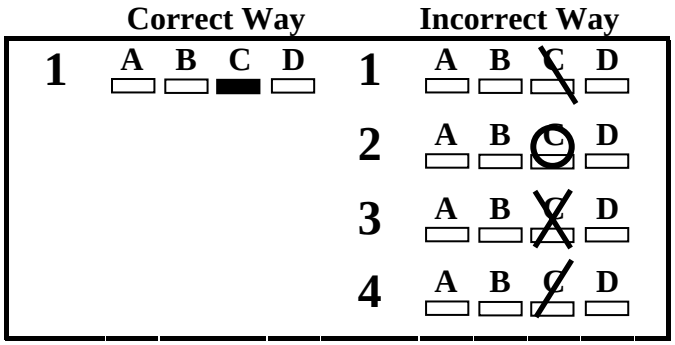
CLASS IX EXAMINATION 2010

Biology Paper I

Time allowed: 35 minutes Marks 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

1. Jelly fish and Hydra are placed in the same phylum i.e. Coelenterata by

   A. ecologists.
   B. biochemists.
   C. taxonomists.
   D. physiologists.

2. The bending and straightening of a human arm follows the principle of leverage. Which branch of science seems to be related to biology here?

   A. Physics
   B. Economics
   C. Geography
   D. Mathematics

3. Which of the following measures controls the increasing rate of pollution?

   A. Introducing new fishing methods
   B. Selective breeding of animals
   C. Controlled use of fertilizers
   D. Preventing abuses of drugs

4. When a hypothesis is developed, it is

   A. directly tested through experiments.
   B. universally accepted by scientists.
   C. proved through gathered observations.
   D. used to form deductions.

5. According to the five kingdom classification system a cell with cellulose cell wall can be placed in

   A. Plantae and Animalia.
   B. Animalia and Fungi.
   C. Protista and Plantae.
   D. Fungi and Protista.

6. Which of the following is NOT a characteristic feature of fungi?

   A. Cell wall is made up of lignin
   B. Absorptive mode of nutrition
   C. Body is composed of hyphae
   D. Act as decomposers in nature

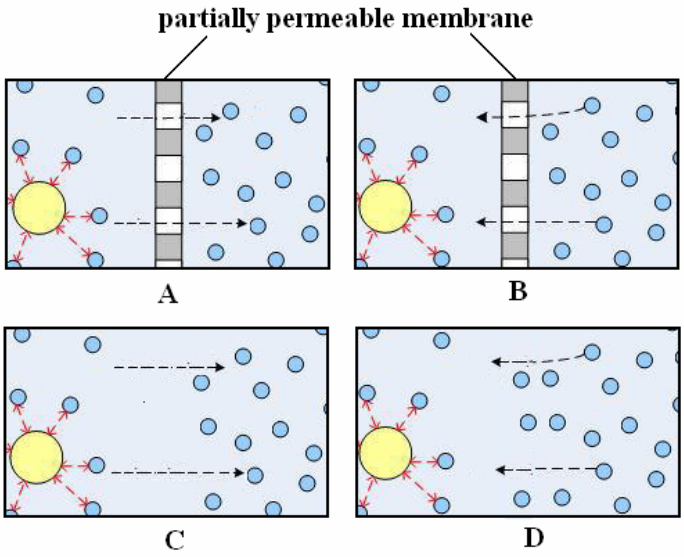
7. Which is the correct representation of naming an organism according to binomial nomenclature?

   A. Amoeba Proteus
   B. Amoeba proteus
   C. amoeba proteus
   D. amoeba Proteus

8. Lysosomes are most abundant in those cells which perform phagocytic activity. Which of the following cells has more lysosomes?

   A. Nerve cell
   B. Epithelial cell
   C. White blood cell
   D. Islets of Langerhans

9. Which of the following diagrams best represents the movement of molecules in osmosis?



10. An animal cell is placed in 20% sucrose solution. What will happen to the cell after some time?

A. Remain unchanged
B. Decrease in size
C. Increase in size
D. Burst

11. A cell has lost its ability to store secretions and convert them into finished products.

In which of the following structures is this defect located?

A. Golgi bodies
B. Mitochondria
C. Cell membrane
D. Endoplasmic reticulum

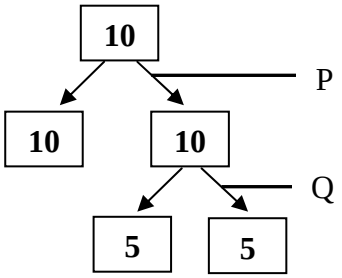
12. During meiosis the process of crossing over takes place in

A. prophase I.
B. prophase II.
C. metaphase I.
D. metaphase II.

13. Which of the following is an example of apoptosis?

A. Destruction of cancerous cells
B. Death of cells by snake venom
C. Premature death of muscle cells
D. Degeneration of red blood cells after 120 days

14. The diagram shows cell division.



Identify the types of cell division taking place at P and Q.

	P	Q
A	Mitosis	Meiosis
B	Mitosis	Mitosis
C	Meiosis	Mitosis
D	Meiosis	Meiosis

15. ATP is known as a "high energy" compound because

A. its one molecule consists of four phosphate groups.
B. its one molecule produces 4.5 kilocalories.
C. its phosphate groups are easily removed.
D. it has strong hydrogen bonding.

16. All the oxygen released during photosynthesis comes from the

A. conversion of starch into glucose.
B. splitting of water molecules.
C. reduction of carbon dioxide.
D. break down of glucose.

17. Which of the following features is NOT an adaptation of leaves to perform photosynthesis?

A. Larger number of stomata
B. Large leaf surface area
C. Waxy cuticle on the leaf
D. High concentration of chlorophyll

18. What happens to an enzyme after it has reacted with a substrate?

A. It is denatured.
B. It remains unchanged.
C. It is broken into products.
D. It requires a co-factor for next reaction.

19. Which of the following is TRUE about anaerobic respiration in humans?

	SITE	END PRODUCT
A	Mitochondria	Lactic acid
B	Mitochondria	Ethanol
C	Cytoplasm	Ethanol
D	Cytoplasm	Lactic acid

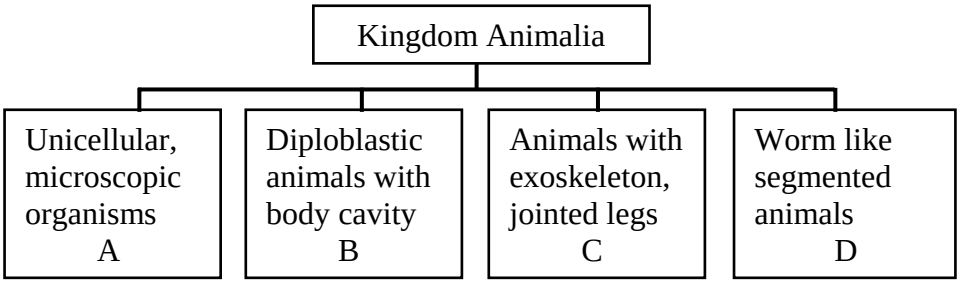
20. Which of the following describes the function of proteins in the body?

A. Solvent for some vitamins
B. Instant energy source
C. Insulating material
D. Growth and repair

21. What should a person suffering with bleeding gums and poor healing of wounds include in his diet?

A. Dairy products
B. Citrus fruits
C. Red meat
D. Cereals

22. Which of the following characteristics indicate phylum Annelida?



23. The movement of water from roots to all parts of the plant is carried out by all of the following EXCEPT

A. root pressure.
B. capillary action.
C. transpiration pull.
D. mass flow mechanism.

24. Which statement is FALSE about haemoglobin?

A. It transports oxygen.
B. It is found in red blood cells.
C. Its deficiency causes leukaemia.
D. It consists of iron and proteins.

25. The blood vessel carrying deoxygenated blood to the lungs is the

A. pulmonary artery.
B. pulmonary vein.
C. coronary artery.
D. coronary vein.

END OF PAPER